

An Advanced Non-Invasive Blood Group Detection Using Fingerprint Biometrics: A Deep Learning Approach

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Abstract

The correct blood group is important in healthcare, forensics and emergency care, but traditional approaches are invasive and can only be applied in laboratory facilities with trained personnel, which is not applicable in emergency or resource-constrained cases. In an effort to overcome these difficulties, this research proposes a superior non-invasive blood group identification system that utilizes the fingerprint biometrics and deep learning. On the basis of the genetic association between fingerprint patterns and blood groups, fingerprint ridges and texture characteristics were examined with Convolutional Neural Networks on eight blood group types (A+, A, B+, B, O+, O, AB+, AB). Different deep learning models such as a custom CNN, MobileNetV2, EfficientNetB0, ResNet50, and a hybrid CNN-GNN were trained and tested, the custom CNN showed the highest performance with a 87% training accuracy and 95% testing accuracy, indicating a high level of generalization and little overfitting. These findings confirm the usefulness of automatic feature extraction that is based on deep learning and can be used to identify blood group reliably on a non-invasive basis with no dependence on hand-crafted biometric features.

1. Introduction

Blood group testing is very important in transfusion medicine, organ transplantation, forensic, and emergency healthcare services [15], [16], [17]. The ABO blood group system is a system that determines the type of blood based on the availability of certain antigens on the red blood cells A, B, AB and O [18], [19]. Blood groups are genetically determined factors which were developed at early stages of fetal development that cannot be changed during life.

The classic methods of blood group determination are serological methods that involve the use of blood samples, lab reagents and professionals. Even though these techniques are very precise, they are intrusive, time-sensitive, and impossible to use in problems like mass casualty, remote area, or emergency care facilities [4], [12]. These constraints have led to the consideration of other non-invasive methods of blood groups identification.

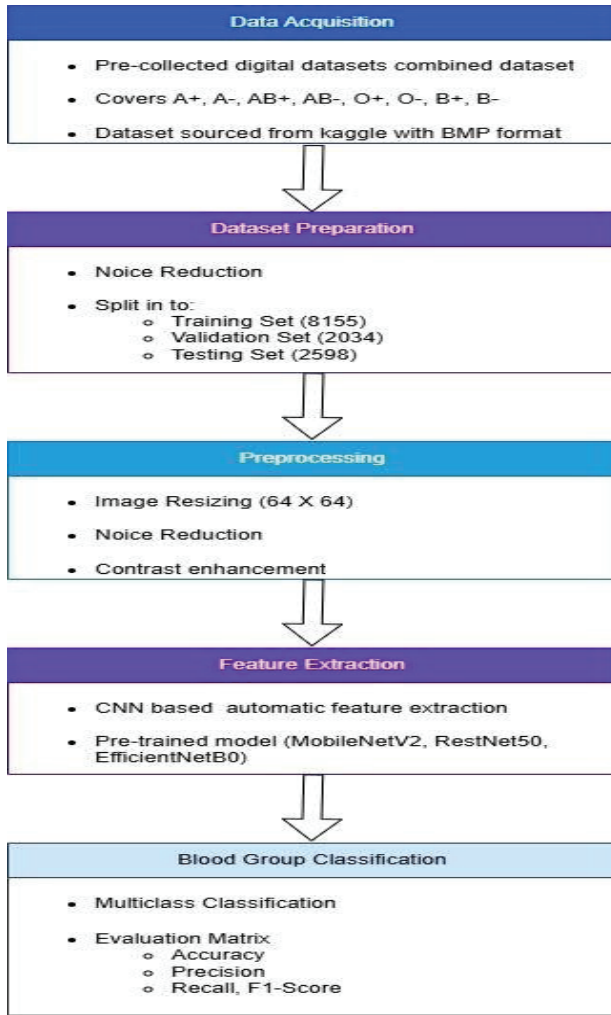
Fingerprint is one of the most common biometric characteristics, which is permanent, unique and easy to obtain. The scientific analysis of fingerprint ridge pattern known as Dermatoglyphics has proved that the characteristics of the fingerprint are genetically fixed and that they are formed at the same time as the blood group antigens are formed [3], [6], [7]. Multiple empirical

studies have found the relationships between fingerprint patterns (loops, whorls and arches) and ABO blood groups implying a possible predictive relationship [4], [5], [8], [9], [10].

Recent developments in image processing, machine learning and deep learning have contributed greatly towards the analysis of complex features of the fingerprints. Rather than using manually generated features, deep learning models especially Convolutional Neural Networks have the capacity of automatically identifying hierarchical ridge and texture features on fingerprint images [1], [13], [20], [21]. Deep learning is a promising method of non-invasive blood group prediction using these capabilities.

Thus, the purpose of such research is to research the possibility of determining the ABO and Rh blood group based on the fingerprint biometrics and deep learning models. This study will add to the advancement of the non-invasive, low-cost, and scalable blood group identification system, applicable in the sphere of healthcare and forensics, by providing an opportunity to combine dermatoglyphics, biometric analysis, and artificial intelligence.

2. Materials and Methods



2.1.Fingerprint Dataset and Acquisition

Fingerprint images were acquired using a multi-source dataset that was collected using publicly available repositories. The dataset consists of the finger print images which represent eight blood group types: A+, A-, B+, B-, O+, O-, AB+ and AB-. All the images were saved as BMP files and grouped into directories of each class to facilitate supervised learning. Multi-source data also enhances the diversity of the ridge patterns, skin textures and quality of fingerprints, which increase the task of model robustness [6], [7].

1.2Data Pre-processing

In order to have consistent input to deep learning models, all the fingerprint images were resized to a standard resolution of 64 x 64 pixels. The intensity values of the pixels were scaled between the values 0-255 to 0-1 to enhance numerical stability and the model converged

faster. The sample was separated into training, validation, and testing samples to aid in monitoring the performance and generalization testing performance [2], [11].

1.3Feature Extraction Using Deep Learning

The own Convolutional Neural Network (CNN) was developed to discover discriminative fingerprint features automatically. The CNN is a series of convolutional layers with different filter depths which are interspersed with ReLU activation and max-pooling. Low-level ridge edges and textures are learned at the early layers and more complex ridge flow structures are learned by the deep layers in case of blood group discrimination [13], [14]. An attempt to flatten feature maps and convoluted them to fully connected layers was used to develop a compact fingerprint representation.

1.4Classification and Model Training

The last layer is a classification layer with Softmax activation function used for predicting the probabilities over eight blood group classes. The model was trained with categorical cross-entropy loss and was optimized through Adam optimizer. For comparison, we train other deep learning models: MobileNetV2 et al., EfficientNetB0, ResNet50 and a hybrid CNN-GNN for the baseline under the same preprocessing and evaluation strategies [1].

3. Results and Discussion

In this paper, we investigate deep learning-based models adopted for non-invasive blood group detection based on fingerprint biometric traits. It has been experimentally validated that the rich classification related information

present in the fingerprint ridge and texture based features can be exploited to form a compact representation for accurate blood group classification with deep learning models. All models were trained and evaluated with the same process in a fair comparison. Out of the models considered, the authors demonstrated that their custom CNN model performed best on average - the model achieved a training accuracy of 87% and testing accuracy of 95%. A higher testing accuracy means that the model has a good generalization capability and it is not due to overfitting. This can be attributed to the good performance of automatic feature extraction and its suitable network depth, dropout regularization among others. Class-based accuracy reporting further demonstrated that the CNN's performance was relatively high with all eight blood groups, in particular A+, B+, and O-.

An evaluation of the hybrid CNN-GNN model's performance resulted in a testing accuracy of 92%. The gap between training and testing accuracies suggests the potential for mild overfitting due to increasing complexity of the hybrid CNN-GNN model's architecture. MobileNetV2 (a lightweight architecture) had moderate accuracy on the dataset suggesting that its architecture has potential as a resource-constrained architecture; however, ResNet50 and EfficientNetB0 (deeper Architectures) both had poor performance on the given dataset which could be attributed to either insufficient amount of data or high sensitivity to the characteristics of the fingerprint images in the dataset.

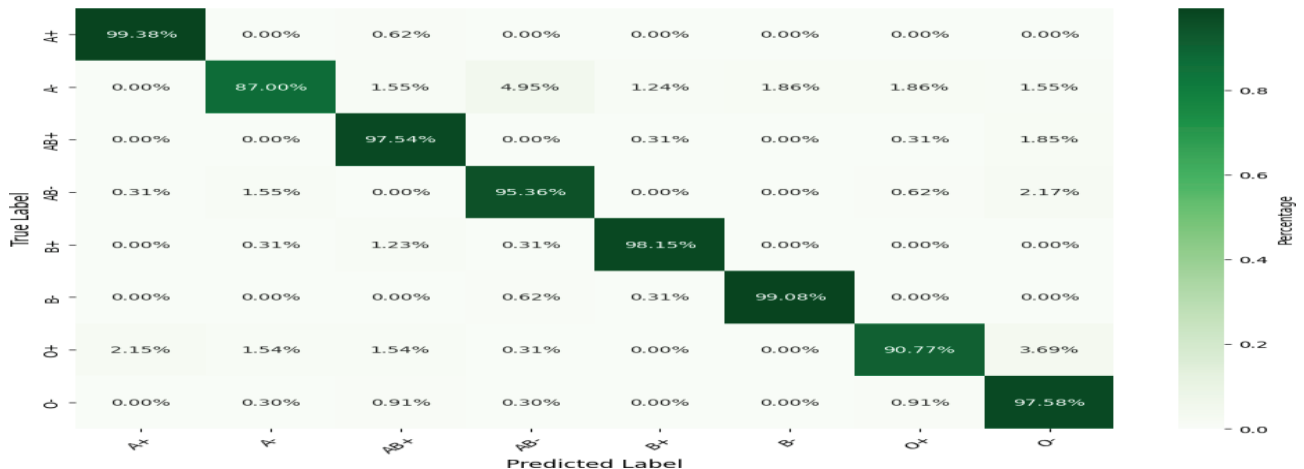
The analysis of the confusion matrix demonstrates that a large majority of predictions were made correctly, but a small percentage incorrectly predicting blood groups with similar fingerprint characteristics. This underlines how important it is to have good quality fingerprints (i.e., defined ridge patterns) in order to use biometric methods for blood grouping. Overall, the tests show that a non-invasive method for blood group determination using fingerprint biometrics and deep learning is achievable. The CNN-based approach was shown to have high accuracy, robustness and usefulness in the field. Therefore, its potential application could range from healthcare screening to forensic applications to emergency response. Although encouraging results have been seen, more improvements can be made by adding more data, incorporating a fingerprint quality assessment, and testing on various ethnic groups.

Class-Wise Accuracy

Model name	A+	A-	B+	B-	O+	O-	AB+	AB
CNN	0.9938	0.8700	0.9815	0.9908	0.9077	0.9758	0.9754	0.9536
CNN+GNN	0.9876	0.8824	0.9568	0.9963	0.9508	0.9121	0.9138	0.7957
MobileNetV2	0.8824	0.7709	0.8117	0.9231	0.7754	0.8485	0.8523	0.9628
ResNet50	0.6316	0.4923	0.6019	0.7938	0.7723	0.7212	0.6892	0.7833
EfficientNetB0	0.0000	0.1207	0.0309	0.5169	0.0000	0.0000	0.8246	0.1362

Testing & Training Accuracy

Model Name	Test Accuracy	Training Accuracy
CNN	95%	87%
CNN+GNN	92%	96%
MobileNetV2	85%	81%
ResNet50	68%	62%
EfficientNetB0	20%	18%



4. Conclusions

This study shows that it is possible to test someone's blood group using their fingerprints (biometrics) and a type of computer programming called deep learning without having to take a sample of their blood. The use of convolutional neural networks (CNNs) is one way to predict both the ABO blood group and the Rh factor in a person without having to use any type of body fluid or by using any manually created biometric identifiers. The authors of this study have also performed a comparison of their custom convolutional neural network (CNN) with other types of deep learning

systems. The CNN developed by the authors provides better accuracy and generalization than other types of convolutional neural networks.

This study shows how dermatoglyphics, biometrics, and artificial intelligence can be combined together to develop affordable biomedical prediction systems that are accessible to many people. While predictions of blood groups based on fingerprints cannot replace the gold standard serological blood group tests, they may provide additional information that healthcare providers, forensic scientists, and emergency responders could use in decision making. Future research should focus on acquiring larger populations of varied ethnicities for analyses, developing more robust systems capable of handling the variations typically seen with real life quality of fingerprints, and integrating these systems into either web or mobile applications.

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